

Bangzhe Huang

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Current position: Research Intern in Prof. Ziming Liu's group, College of AI, Tsinghua University

Research interests: automated research, research world models, AI4AI, scientific machine learning, and many-body physics

Education

Fudan University, Department of Physics

Shanghai, China

Incoming undergraduate, Strong Foundation Program in Physics

September 2026 – expected June 2030

Research Experience

AI4AI: Learning to Design a Model

Lead researcher; advisor: Prof. Ziming Liu, College of AI, Tsinghua University; active pilot

- Built a controlled AI4AI pilot in which a graph meta-model learns the mapping from neural architectures to downstream test log-MAE under fixed regression tasks, data splits, optimizers, and training budgets.
- Defined a DAG architecture DSL from four homogeneous MLP blocks and one softmax gate, exhaustively enumerated 10,006 canonical candidates, and encoded component and edge structure with a graph neural network.
- Developed an interactive demo for manual DAG construction, DSL validation, outcome prediction, automatic architecture design, and training dynamics. Preliminary experiments identified a high-performing ResNet-like design.

Scaling Foundation Models for Many-Body Physics

Project lead; advisor: Prof. Jing Wang, Department of Physics, Fudan University; preprint in preparation

- Identified severe optimization ill-conditioning as a key barrier to scaling a monolithic model across heterogeneous Hamiltonian families, and proposed a task-modular architecture that preserves family-specific physical priors and ansatzes while enabling cross-task knowledge composition.
- Led problem formulation, method design, PyTorch implementation, and evaluation against unified joint-training baselines across task scales, task combinations, and fine-tuning settings.
- Preliminary results show comparable generalization and adaptation when unified training remains tractable, while the modular system remains trainable in regimes where the unified baseline becomes ill-conditioned.

Open-Ended Scientific Research Benchmark for Physics Research Agents

Independent research; academic discussion: Prof. Siheng Chen, School of Artificial Intelligence, Shanghai Jiao Tong University; pilot completed

- Designed open-ended physics research tasks with stable ground truth defined by low-dimensional physical structures and equivalent observables, without prescribing a single valid research trajectory.
- Built an end-to-end pilot spanning experiment and simulation design, code generation and execution, multi-round observation and feedback, and a final physical audit.

Phase Transport and Defect Dynamics in Nonreciprocal Spin Systems

Lead researcher; advisor: Prof. Zi Cai, School of Physics and Astronomy, Shanghai Jiao Tong University; pilot completed

- Used Monte Carlo simulations to study the microscopic origin of macroscopic oscillations in two-dimensional nonreciprocal spin systems and extracted oscillation frequency, domain-wall density, and defect observables.
- Connected oscillation frequency ω , domain-wall density ρ , and effective transport velocity through $\omega = \rho v_{\text{eff}}$, and extracted v_{eff} from the linear ω - ρ regime.
- Identified regimes dominated by vortex-pair generation and periodic wall expansion versus vortex-constrained topological structures, and tested the physical picture across lattice geometries and symmetric bond disorder.

Honors

Gold Medal, 42nd Chinese Physics Olympiad, 2025

Technical Skills

Machine learning: PyTorch, deep learning, scientific foundation models, graph neural networks, LLM agents, benchmark design

Scientific computing: Python, Monte Carlo simulation, numerical experiments, statistical analysis

Physics: quantum mechanics, statistical physics, many-body physics